

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL
ONLINE INFORMATION AND TOURIST NAVIGATION FOR KML MOUNTAIN RESORT

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Francis Elmido – BSIT 3
Jaryll Judicpa – BSIT 3
Junard Calderon – BSIT 3
Jesar Lagunzad - BSIT 3
Earon Janeo – BSIT 3

System Development Track

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The rapid growth of web-based technology has transformed how people discover travel destinations and navigate unfamiliar locations. Online platforms allow resorts to reach guests far beyond their local area, providing essential information and precise directions before a traveler even leaves home.

However, many smaller leisure destinations, such as KML Mountain Resort, face significant challenges in this digital landscape. Despite offering premium amenities like panoramic mountain views and private accommodations, the resort suffers from low online visibility and difficult physical navigation. Currently, the resort's location is hard for search engines to find, and its remote nature often confuses guests. Furthermore, the lack of a centralized digital hub means the owners must manually answer every inquiry, which is inefficient for a small-scale operation.

To address these issues, modern web development offers a targeted solution. By integrating Interactive Map APIs and Search Engine Optimization (SEO), a resort can transform from a hidden location into a highly visible and easy-to-reach destination. An integrated web portal can provide real-time navigation using GPS coordinates and visual landmarks, ensuring that guests are guided directly to the resort gates without confusion.

This capstone project proposes the development of an Online Information and Tourist Navigation System for KML Mountain Resort. The system is designed to provide precise navigation tools, a digital information hub, and increased search engine discoverability. By applying principles of web development, database management, and digital mapping, this project serves as a practical application of the knowledge gained by BSIT students to solve real-world business challenges.

1.2 Statement of the Problem

KML Mountain Resort suffers from low online visibility and difficult physical navigation due to its remote location, which hinders its ability to reach its target market and results in lost potential revenue and operational inefficiencies.

1.Low Search Engine Discoverability: The resort lacks an official SEO-optimized digital platform that makes it difficult for city-based and foreign tourists to find the business through organic online searches.

2.Confusing Navigations: Because the resort is "secluded" and remote the standard GPS tools often fail to provide the precise route leading to guest frustration and difficulty in locating the entrance.

3. Manual Information Dissemination: There is no centralized digital gallery or information hub, forcing the owners to manually handle repetitive inquiries regarding room arrangements, food services, and resort facilities.

4. Limited Brand Awareness: Without a professional web presence, the resort struggles to compete with more visible destinations, limiting its ability to attract event organizers and families looking for a secure getaway.

1.3 Objectives of the Study

1.3.1 General Objectives

The primary goal of this study is to design and develop an Online Information and Tourist Navigation System for KML Mountain Resort that enhances the resort's digital visibility and provides precise, landmark-based navigation for guests.

1.3.2 Specific Objectives

Specifically, the study aims to:

- To design a web-based portal that serves as the official digital information hub, showcasing the resort's facilities, services, and digital gallery to set guest expectations.
- To implement an Interactive Map Integration using GPS coordinates and visual landmarks to guide tourists directly to the resort gates through their mobile GPS applications.
- To streamline the user flow for potential guests by providing a seamless transition from discovering the resort online to initiating turn-by-turn navigation.
- To reduce manual administrative tasks for the owners by automating the dissemination of resort information and frequently asked questions.

1.4 Scope Delimitation

The study focuses on the development of an online information and tourist navigation portal for KML Mountain Resort. The system will include a digital information hub, a high-quality photo gallery of facilities, an interactive location module with landmark-based guidance, and an administrative dashboard for content management.

The navigation component will utilize GPS coordinate integration and external mapping links to provide precise, turn-by-turn directions to the resort gates. The study will not cover integrated online payment processing, real-time room inventory synchronization with third-party booking sites, or offline map functionality. The application will be developed as a responsive web-based

prototype for academic purposes, designed to be accessed by both administrators (resort owners) and end-users (tourists) via mobile and desktop browsers.

1.5 Significance of the Study

The proposed system will benefit the following:

- Tourists and Guests** – by providing easy access to resort information and precise, stress-free navigation to the secluded location.
- Resort Owners** – by reducing manual inquiries, increasing business visibility, and providing a professional platform to showcase their services.
- Local Tourism Industry** – by promoting "hidden" local destinations and demonstrating how technology can boost the reach of small-scale mountain resorts.
- BSIT Students** – by serving as a practical application for learning web development, API integration (Google Maps/Waze), and Search Engine Optimization (SEO).
- Future Researchers** – by providing a reference for studies related to location-based services, digital tourism, and the digital transformation of remote small businesses.

1.6 Definition of Terms

- API (Application Programming Interface)** – A set of tools that allows your website to "talk" to another service, such as linking your site to Google Maps for navigation.
- Digital Gallery** – A specific section of the website used to display high-quality photographs of the resort's rooms, pool, and views.
- GPS (Global Positioning System)** – The satellite-based service used by mobile devices to provide the exact coordinates and location of the resort.
- Integrated Resort Web Portal** – A centralized website that combines different features like information sharing, photo galleries, and navigation tools in one place.
- Interactive Map** – A digital map on the website that users can zoom into, move around, and click on to get live directions.
- Landmarks** – Easily recognizable physical objects (like a specific bridge or a large tree) used in the navigation system to help guests know they are on the right path.
- Responsive Web Design** – A way of building a website so that it looks good and works perfectly on both a large computer screen and a small smartphone.
- SEO (Search Engine Optimization)** – The process of setting up the website so that it shows up higher in the list of results when someone searches for "resort" on Google.
- User Flow** – The path a guest takes through the website, from the moment they land on the home page to the moment they click "Navigate" to start their trip.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Miller and Thompson (2021) discussed the application of Integrated Information Systems in the hospitality industry. Their study revealed that web-based portals improved business discoverability by centralizing resort amenities and services into a single accessible platform. Similarly, Chen et al. (2020) emphasized that intelligent navigation systems significantly enhance user satisfaction and reduce travel anxiety for tourists visiting remote or "hidden" destinations.

2.2 Local Studies

According to Bautista and Mendoza (2022), application-based systems developed in local academic settings improve the promotion of provincial tourism and data accuracy for small businesses. Their study on location-based web applications highlights the importance of user-centered design and precise geographic mapping. Garcia (2021) also emphasized that integrating real-time navigation tools enhances the reliability and usability of local resort websites, making them more competitive in the digital market.

2.3 Related Literature

Geographic Information Systems (GIS) literature explains that coordinate-based mapping uses latitude and longitude data to provide precise location tracking (Longley, 2020). In application development, Pressman (2019) stressed the importance of structured software engineering methodologies to ensure system quality and cross-platform compatibility. Studies on digital tourism also highlight discoverability and ease of navigation as major factors in a guest's decision to book a remote resort (Werthner et al., 2018).

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies demonstrate that integrated web systems significantly improve business visibility and guest navigation. However, limited research focuses specifically on boutique mountain resorts that require a combination of landmark-based navigation and administrative information hubs. This study addresses this gap by integrating

GPS-based tourist navigation with web application development, aligning with current technological and academic trends in the Philippine hospitality sector.

References

Bautista, J. & Mendoza, L. (2022). Location-Based Web Applications for Provincial Tourism. Philippine Journal of IT Innovations.

Chen, H., et al. (2020). Navigation Systems and Guest Satisfaction in Secluded Destinations. International Journal of Hospitality Tech.

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Longley, P. A. (2020). Geographic Information Systems and Science. Wiley & Sons.

Miller, R. & Thompson, S. (2021). Integrated Information Systems in the Modern Hospitality Industry. Journal of Business and Tech.

Pressman, R. S. (2019). Software Engineering: A Practitioner's Approach. McGraw-Hill.

Werthner, H., et al. (2018). Digital Tourism: Discoverability and Decision-Making. Springer Tourism Series.

CHAPTER III METHODOLOGY

3.1 Research Design

The study will use a developmental research design, focusing on the design, development, and evaluation of the proposed Online Information and Tourist Navigation System for KML Mountain Resort.

3.2 System Development Methodology

The **Agile Development Model** will be used, consisting of the following phases:

1. Requirements Analysis: Identifying the specific needs of KML Mountain Resort, such as landmark-based navigation and photo gallery requirements.
2. System Design: Drafting the user interface and planning the integration of the Google Maps API.
3. Development: Coding the web portal and setting up the administrative dashboard.
4. Testing: Conducting trials to ensure navigation accuracy and website responsiveness.
5. Deployment (Prototype): Presenting the working web portal for the resort owners and guests to use.

3.3 System Architecture

The system will follow a client-server architecture. Users will interact through a responsive web interface (mobile or desktop), while the server will handle data processing, map coordinate retrieval, and database management for resort information.

3.4 Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js/Express.js
- Mapping Integration: OpenStreetMap
- Database: MySQL
- Development tools: Visual Studio Code

3.5 Data Gathering Techniques

- Surveys and questionnaires provided to potential tourists and guests.
- Interviews with the resort owner, Michelle Tingson Miranda, to identify navigation pain points.
- Review of existing tourism platforms and maps to identify information gaps.

3.6 Testing Procedures

- Unit Testing: Verifying those individual components such as the navigation buttons, function correctly.
- System Testing: Testing the end-to-end process from landing on the home page to launching the GPS app.
- User Acceptance Testing: Gathering feedback from the resort owners and guests to ensure the system meets their real-world needs.

3.7 Evaluation Criteria

- Functionality: Does the system successfully provide resort info and directions?
- Usability: How easily can a guest find and use the navigation features?
- Accuracy of Navigation: Does the system reliably guide the user to the correct resort entrance?
- Security: Is the owner's administrative panel protected from unauthorized access?
- Efficiency: Does the portal load quickly on mobile devices with varying signal strengths?

SUPPORTING DIAGRAMS

3.8.DATA FLOW DIAGRAM (DFD - Level 0)

This represents the high-level flow of information between the users and system.

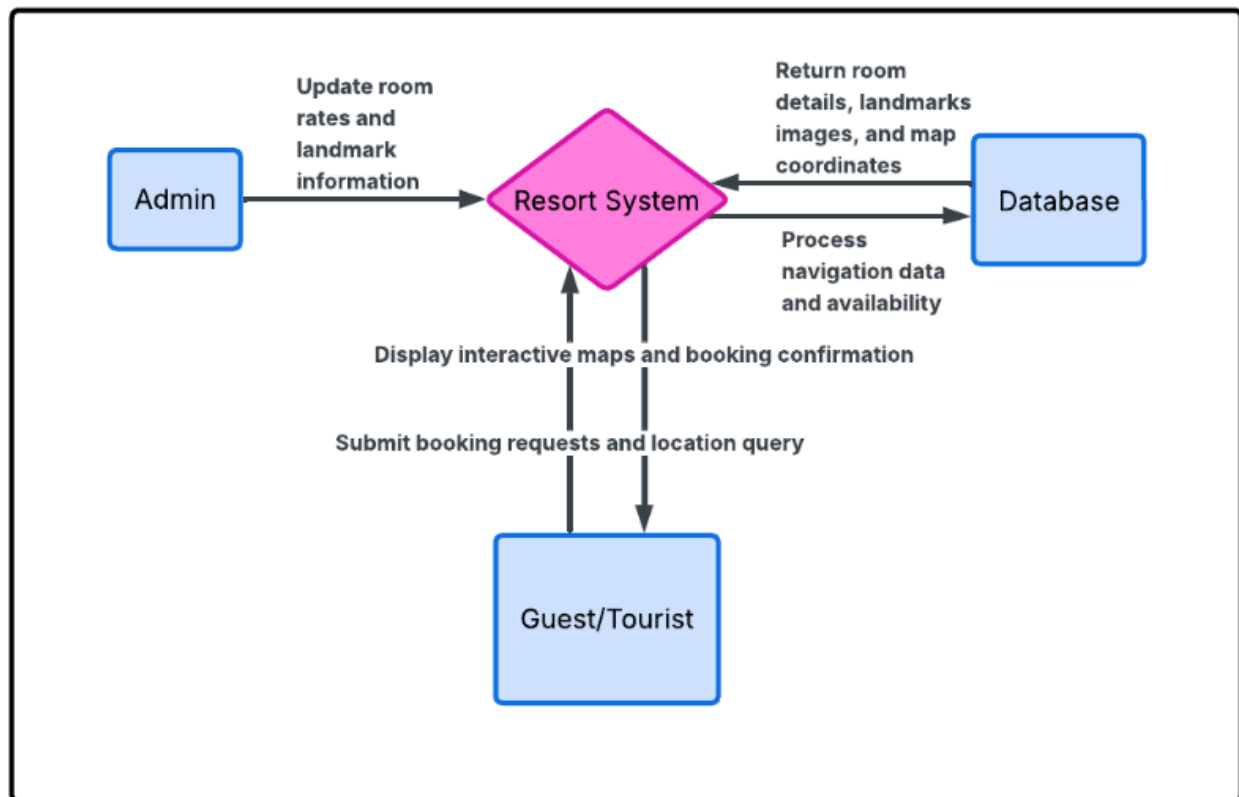
Guest/Tourist → Submit booking requests and location query → **Resort System**

Resort System → Process navigation data and availability → **Database**

Database → Return room details, Landmarks images and Map Coordinates → **Resort System**

Resort System → Display Interactive Maps and Booking Confirmation → **Guest/Tourist**

Admin → Update room rates and Landmark Information → **Resort System**



3.9 ENTITY RELATIONSHIP DIAGRAM (ERD)

Entities And Attributes:

Guest (Guest ID, Name, Email, Phone)

Booking (Booking ID, Guest ID, RoomType, CheckIn, CheckOut, Status)

LandMark/Map (PointID, Coordinate, Description, ImageURL)

Admin (AdminID, Username, Password, Role)

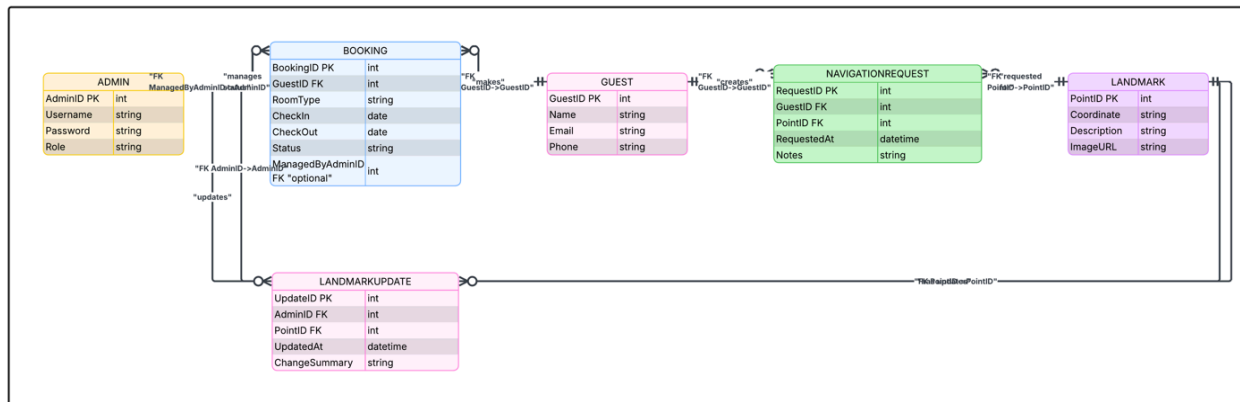
Relationships:

Guest make a Booking.

Admin manages Booking Statuses.

System provides Landmark data to the guest for Navigation.

Admin updates Landmark and Resort Information.



3.10 USE CASE DIAGRAM

Actors:

Tourist/Guest - The end-user seeking Information and directions.

Administrator - The resort owner, Michelle Tingson Miranda, managing the portal.

Uses Case:

View Resort Information - Browse the digital gallery and amenities.

Inquire/Book Room - Submit the reservation form.

Access Live Navigation - Use the Interactive Map with landmark guidance.

Manage Content (Admin) - Update the photo gallery and resort details.

Approve Bookings (Admin) - Review and update the status of guest requests.

Monitor Analytics (Admin) - See how many users are accessing the navigation tools.

